

CEPHALOPOD CARTOGRAPHIES: AN INVESTIGATION OF SPATIAL
COGNITION AND NAVIGATION IN THE SEPIOLID *EUPRYMNA BERRYI*

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PROJECT SUMMARY

Cephalopods are a unique class of mollusks that have undergone remarkable evolutionary changes, driven by a long-standing arms race with marine fishes since the Devonian period. This pressure has led to a suite of advanced adaptations, including a high degree of cephalization reflected in their large and complex brains. Cephalopods possess some of the highest brain-to-body mass ratios among invertebrates, even rivaling many vertebrates. Their neural sophistication, along with diverse behavioral repertoires observed in both natural and laboratory settings, has made them a compelling subject for cognitive research. Since the 1960's scientists have explored cephalopod learning, memory, problem-solving, social behavior and communication, and navigation, with most work focusing on octopus and cuttlefish models. This thesis investigates spatial and navigational cognition in a lesser-studied lineage: the sepiolids. Members of the family Sepiolidae are small, benthic cephalopods closely related to cuttlefish, yet their cognitive capabilities remain poorly understood. The present study focuses on *Euprymna berryi*, a sepiolid native to Indo-Pacific waters, and explores its spatial navigation abilities using a circular open-field tank. The experimental design tests multiple forms of navigation: initial and successive fluent navigation of novel environments, use of external tank cues such as polarized light, path integration, piloting based off landmarks, and navigation through beaconing. By assessing the strategies employed by *E. berryi* under different conditions, this work aims to expand our understanding of spatial cognition in cephalopods. These findings will not only contribute to the growing field of invertebrate cognition but also offer broader insights into the evolution and diversity of animal navigation strategies.

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PROJECT DESCRIPTION

Background and Significance

Cephalopods are a unique and complex group of invertebrates that have fascinated scientists and the public alike for decades. This class of mollusks is famous for their complex nervous system, consisting of an advanced brain and eyes that are convergent with those found in vertebrate animals. The complexity of these characteristics are thought to have developed and diversified during the Devonian Period when competition with jawed fishes became unavoidable: cephalopods, being both predator and prey, needed to develop attributes necessary to stay in competition with fish for prey items and to avoid becoming prey items in the process themselves. Instead of hindering the cephalopods, this 400 million-year-old ecological predicament led this specific class of invertebrates to evolve brain and ocular structures convergent to those of their vertebrate competitors (Land and Nilsson 2002, Packard 1972).

Extensive work on mapping the cephalopod brain has revealed that although it is morphologically dissimilar to vertebrate brains, it is composed of regions termed “lobes”, such as the the vertical, superior and inferior frontal, and optic lobes, that each handle different types of cognitive information which and are analogous to certain areas in vertebrate brains (Hochner et al 2006). The vertical lobe is the center for associative processing, learning, and memory capabilities; it is analogous to the vertebrate hippocampus and neocortex (Shomrat et al 2008, Fiorito and Chichery 1995, Young 1991, Maddock and Young 1987). The superior frontal lobe is responsible for sensory and processing input into the vertical lobe, like the vertebrate prefrontal cortex and sensory association cortices (Schnell et al 2021, Young 1991, Maddock and Young 1987). The inferior frontal lobe deals with tactile learning through the use of their arms and suckers, which can be likened to the vertebrate somatosensory and motor cortices

(Schnell et al 2021, Young 1991, Maddock and Young 1987). As the final but largest regions the optic lobes are utilized for the processing of visual information, object recognition, and motion detection; this area shares similarities to the vertebrate primary visual cortex and superior colliculus (Budelmann 1995, Maddock and Young 1987). The lobes mentioned are just a small portion of total brain mass; octopuses alone were discovered to have over 30 specialized brain lobes (Styfals 2022). All of these specialized regions come together to create an advanced and specialized brain, the brains in cephalopods are fairly large for their size, having this group of animals exhibit high brain-to-body ratios among the rest of their invertebrate peers, even being comparable to those of some vertebrates (Packard, 1972).

Cephalopod eyes are an even better example of a nervous system structure that is convergent with vertebrates: both are described as camera lens-type eyes (save for Nautilus species, which have a pinhole-type eye) that have varying refractive indices, are prominent and large, and are capable of manipulating pupil morphology (Land and Nilsson 2002, Packard 1972). However, there are a fair amount of differences between cephalopod and vertebrate eyes, like the absence of color vision, though this factor seems to have no hindrance on these animals' incredibly accurate camouflaging capabilities with their environments (Marshall and Messenger 1996). Also, cephalopods do not utilize the retina for visual processing like in vertebrates, but instead have that information be processed in the optic lobes of their brain (Williamson and Chrachri 2004).

The advanced capabilities of cephalopod brains and eyes without a doubt aid these animals in many facets of their natural lives, despite those very lifespans being comparatively short to many other highly cephalized animals. Nevertheless octopuses, squids, cuttlefishes, and even the more primitive nautilus all have demonstrated many different facets of their cognitive and visual capabilities through behaviors seen

observationally or tested experimentally. Problem-solving, various kinds of learning, short and long-term memory, foraging strategies, predator avoidance and shelter acquisition, sociality and mating techniques, and cognitive mapping/spatial navigation (Darmaillacq et al 2014). In fact, this last point is very important, as spatial navigation and the generation of a cognitive map of your surrounding environment is intertwined with many of the aforementioned behaviors and is therefore a key component in the day-to-day survival of all cephalopod species.

The spatial and navigational capabilities of cephalopods have been investigated multiple times across a handful of cephalopod groups, though it is primarily looked at in octopuses and cuttlefishes as these two groups are the most suited for laboratory conditions/studies and can be found fairly readily off of coastlines for in situ environmental observations. These studies have yielded some great insights into cephalopod navigation; octopuses have been found to have recollection memory of past dens (Mather 1991), cuttlefishes can switch between place strategy (using surrounding/proximal environmental cues for orientation) and response strategy (left/right positioning for orientation) when exiting a tank maze (Alves et al 2007). Both octopuses and cuttlefish have demonstrated the capability for conditional discrimination likely used in spatial navigation (Hvornecy et al 2007), and interestingly this has also been seen in a more primitive cephalopod group, the nautilus, which were seen to utilize both proximal and distal navigational cues and exhibited a plasticity in navigation when the testing environment was manipulated (Crook et al 2009, Crook and Basil 2013). These studies all serve as a great foundation for understanding navigation, spatial learning and memory, and cognitive mapping in cephalopods, though there is still a lot of headway to make in this area of cephalopod cognition. There are many other cephalopod groups that have yet to be looked at in terms of their navigational abilities:

specifically many of the deepwater cephalopods, true squid species, and the bobtail squids.

The latter of these underexplored groups, family Sepiolidae, presents a new model for studying navigation in cephalopods. Though they are more commonly known as the bobtail squids, the sepiolids are considered more closely related to cuttlefish than they are to the true squids. The intricacies of their phylogenetic placements among the cuttlefishes is debated; they are often placed under the cuttlefish order Sepiida and suborder Sepiolida, or are classified as their own order Sepiolida (Sanchez et al 2021). Bobtails are small and compact in morphology with male mantle lengths ranging from about 1 to 8 cm and can be found in a variety of habitats worldwide in both shallow and deep water, primarily occupying benthic communities as both a predator and prey item. Like many other species of cephalopods, many members of Sepiolidae are nocturnal, choosing to spend the daytime buried within the substrate of its environment and leaving its refugia at night for quick bouts of foraging for prey or to breed (Drerup 2022). Over the past few decades many species of sepiolids have proven to be readily culturable in laboratory conditions- their small size and toleration of group living lends them to be housed fairly easily in the laboratory. They also do not have a larval stage like many other cephalopods as they hatch as tiny versions of the adults and have been cited to be quite “precocious” in behavior post-hatching, with some being able to attack small mysid shrimp within a day of leaving the egg (Jolly et al 2022).

Scientific interest in bobtail squids is generally concentrated towards the symbiotic relationship some species have with the bioluminescent bacteria *Vibrio fischeri*, which assimilates into the light organ of hatchlings and provides the bobtails with counterillumination against downwelling moonlight once it grows. This counterillumination allows the bobtails to become almost invisible to anything below it, obscuring it from any potential prey or predators (Visick and McFall-Ngai 2000). Because

this symbioses serves as one of the main scientific focuses for sepiolids, studies on their behavior and cognition are lacking in comparison to other cephalopod groups. Of the few that have taken a look at possible cognition/behavior in this cohort, one study utilizing an associative learning procedure called the “Prawn-In-The-Tube Procedure” was the first to document stable rapid and long term memory capabilities in the species *Euprymna scolopes*, the Hawaiian bobtail. This procedure classically has a live prey item (a shrimp or prawn) inside a clear glass tube or beaker given to individual cuttlefish, then the prey response behavior of the cuttlefish is observed with the cuttlefishes inhibiting their striking behavior and learning that they cannot get to the prey in the tube. When *E. scolopes* was tested under this procedure, they exhibited a rapid inhibition of their prey response over repeated trials and showed evidence for both short and long term memory- up to 4 and 12 days after the initial training trials (Zepeda et al 2017). This long term memory was directly compared to data gathered in a cuttlefish procedure which was 24 hours, obviously and surprisingly lower than what the bobtails achieved (Agin et al 2003). Learning and memory, as stated previously, is largely a function of the cephalopod vertical lobe. Cuttlefish allocate more than twice the brain volume to their vertical lobes compared to bobtails, whereas the latter slightly surpass cuttlefish in the volume of the optic and inferior frontal lobes—regions associated with visual and tactile processing, respectively (Ponte et al 2021). These differences, especially in the vertical lobe, suggest that sepiolids may have the capacity for spatial navigation and cognitive mapping.

In order to discuss the intricacies of animal navigation and cognitive mapping, it is important to define some commonly used terminology. Orientation refers to directional movements and are not thought to involve cognition. Cognition, in this sense, is defined as “any adaptive (changing with experience) process which connects sensing in the world to action in the world,” or simply the processing of information on your

surroundings (Darmaillacq et al 2014). Navigation is defined as the “process of determining or maintaining a course or trajectory to a goal location” (Franz and Mallot 2000). Combining navigation with cognition produces the term “cognitive mapping” wherein an animal has the capability to create a mental representation of the layout of their spatial environment and have an understanding of where they are in relation to surrounding cues or landmarks due to habituation to that environment over time.

Cognitive mapping and subsequent navigation, when compared to orientation, are more cognitively-involved processes as there can be many different techniques that animals can utilize. Generally, some of the main navigational processes in animals consist of path integration, piloting, and beaconing. Path integration is “the reckoning of a new position and heading from an old position by summing successive intervening displacements and changes in heading to obtain the net change in position and heading” (Gallistel and Matzel 2013). This type of navigation is unique from the rest in that it does not have the animal rely on surrounding environmental cues; path integration is navigation solely generated by the animal internally and based on its prior movements. Conversely, both beaconing and piloting navigational techniques require the utilization of surrounding environmental cues. Beaconing is a simpler navigational technique and requires a lesser cognitive load; animals dictate their movement towards a singular distant landmark (Cheng and Spetch 1998). Piloting is a mechanism on the opposite side of the cognitive load spectrum because it encompasses directionalized, purposeful movement by having an understanding of the spatial arrangement of an environment and where an animal is oriented to physical objects and features (Gallistel 1993). Environmental cues and landmarks can also be made more apparent and useful when highlighted by the polarization of light, which is something all cephalopod photoreceptors are capable of detecting (Mäthger et al 2009). This array of navigational strategies

provides a great template for looking into the possible ways sepiolid cephalopods get around their habitats.

Hypotheses

I hypothesize that sepiolids are capable of spatial navigation and cognitive mapping and utilize piloting, beaconing, and/or path integration. I predict that my model species, *Euprymna berryi*, will successfully navigate the experimental tank to locate the escape. Furthermore, I expect their navigational efficiency—reflected by reduced escape times and/or more streamlined or shorter paths—to improve across repeated trials in all procedures.

Project Methods

Embryos of *Euprymna berryi* (n= 30) were bought from the Marine Biological Laboratory at Wood's Hole, Massachusetts from the MBL Cephalopod Breeding Program and reared in the lab. On the main campus I reared the *E berryi* in 10-gallon glass aquaria equipped with a hang-on-back filter, bubble stone and air pump for aeration, and an aquarium heater connected to an Open Acidification Tank Controller to control temperature. Seawater is made artificially by running tap water through a reverse-osmosis/deionization (RO/DI) filtration system and then mixed with Instant Ocean brand artificial sea salt to achieve a salinity of ~33 ppt. At Rosario Beach Marine Laboratory my seawater system will be a 515 L (136 gallon) closed marine system of recirculating seawater. A 57 gallon (215 L) tank acts as a sump system and is fitted with a protein skimmer, a bubbler, heater, and a canister filter; water is pumped from the sump system into five 20 gallon (75 L ea) acrylic tanks. Squids are fed live grass shrimp or other comparable marine shrimp twice a day and are housed in groups of 2-3 individuals.

The experimental open-field tank (similar to Fig. 1, Crook et al 2009) will be a round section of septic tank riser piping (60 cm diameter, 10 cm deep) with opaque sides and a clear acrylic bottom. One hole is drilled into the acrylic tank bottom on one end of the tank near the wall ($8\frac{1}{2}$ - 10 cm diameter, 3 cm from the tank wall) for the squid to swim down into a holding tank with preferred substrate. The water level in the experimental tank is kept at around 10-12 cm deep, and has no substrate to encourage the squid to escape to the holding tank below. Inside the experimental tank various rocks and plants are arranged around the tank, while a specific marker (one ping pong ball tethered to a small weight by fishing line) floats just behind the exit hole. This entire rounded experimental tank sits on top of a tank with substrate, both of which sit within a larger tank/bin. Data measurements are collected via GoPro camera (sitting 60 cm above the experimental tank), with path length and directionality analyzed through ImageJ.

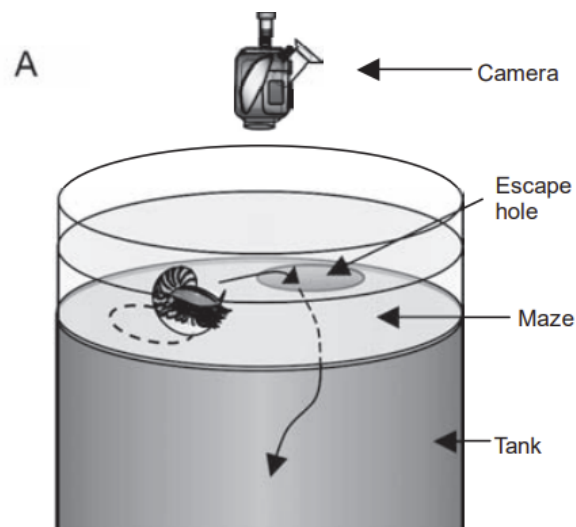


Fig. 1

Trial Arrangements/General Procedures, and Timeline of Use at Rosario

Each squid (n= 10, adults) will be put through five 10-minute trials per procedure type (5 procedures, termed P1-5, outlined below) for a total of 50 trials per procedure type. Two different quantitative types of data will be collected: the time it takes for the bobtail to escape the experimental tank in seconds, and the path distance from the starting position to the escape hole. Each individual is placed at the designated starting position in the open-field experimental tank and have a total of 10 minutes to escape the maze; after 10 minutes have elapsed, subjects are removed regardless of escape, then replaced to be tested again. In an effort to mitigate stress and to provide an escape “reward,” the subjects are given 5 minutes to acclimate in between trials.

Each procedure is expected to take a minimum of 12.5 hours (1 trial takes a minimum of 15 minutes, 5 trials takes ~75 minutes, 75 minutes x 10 squid = 750 minutes or 12.5 hours). 3 days will be allocated per procedure testing 3 groups of squids: Group 1 (squids 1-4), Group 2: (squids 5-8), and Group 3: (squids 9 and 10). This arrangement allows an easier way of keeping track of which subjects have been tested (they are housed in twos). Additionally, the trials are spaced into manageable durations, allowing room for any necessary tweaks or adjustments to the procedures and gives the subjects time off from experimentation in an alternating fashion which aids in reducing stress. The procedural timeline has been written in the calendar below:

June

1	2	3	4	5	6	7
8	9	10	11	12	13	14

15	16	17 Truck Arrives	18 Establish Squids ->	19 ->	20 ->	21
22	23 Dive Checkout ->	24 ->	25 ->	26 ->	27 ->	28
29	30 Experimen tal Tank ->					

July

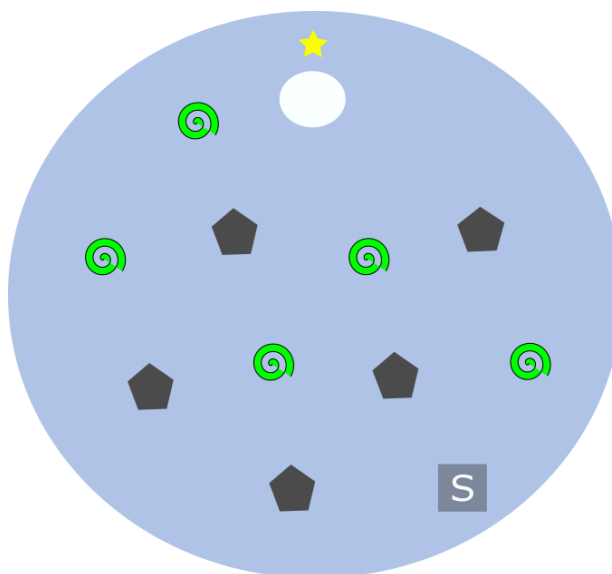
		1 ->	2 ->	3 ->	4 ->	5
6	7 P1: Squids 1-4	8 P1: Squids 5-8	9 P1: Squids 9&10	10 P2: Squids 1-4	11 P2: Squids 5-8	12
13	14 P2: Squids 9&10	15 P3: Squids 1-4	16 P3: Squids 5-8	17 P3: Squids 9&10	18 P4: Squids 1-4	19
20	21 P4: Squids 5-8	22 P4: Squids 9&10	23 P5: Squids 1-4	24 P5: Squids 5-8	25 P5: Squids 9&10	26
27	28 Room for trouble- shooting ->	29 ->	30 ->	31 ->		

August

					1	2
3	4	5	6	7	8	9
10	11	12	13	14 Classes End	15	16

Experimental Procedure 1 (Initial Navigation and Escape of a Novel Environment)

In this preliminary procedure, bobtails are placed in the designated starting position 135 degrees from the exit hole:

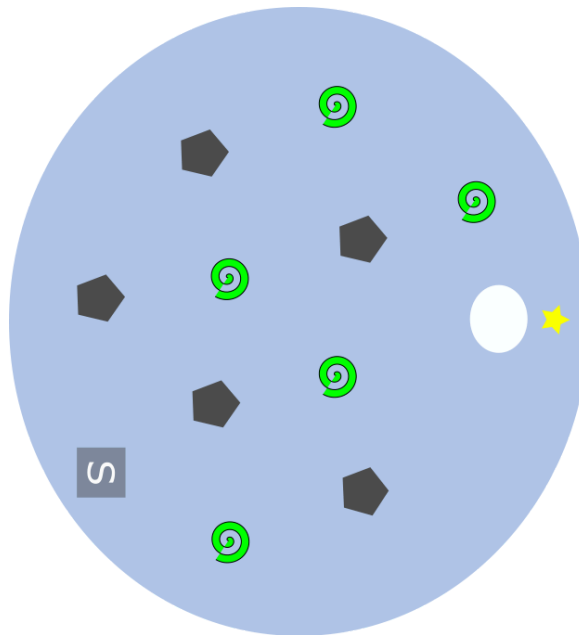


Escaping through the hole tells us they are capable of navigating novel environments; decreasing escape time and/or path distance would show them learning to navigate to this hole over successive trials. A decrease and subsequent stabilization in the time it takes to escape the tank over repeated trials would indicate that the squid is learning and remembering the hole's location using either internal cues, external cues, or

both, each of which will be tested in the subsequent procedures. Similarly, a decrease and subsequent stabilization in path distance to the hole shows us the same conclusion.

Experimental Procedure 2 (Investigation of External Tank Cues)

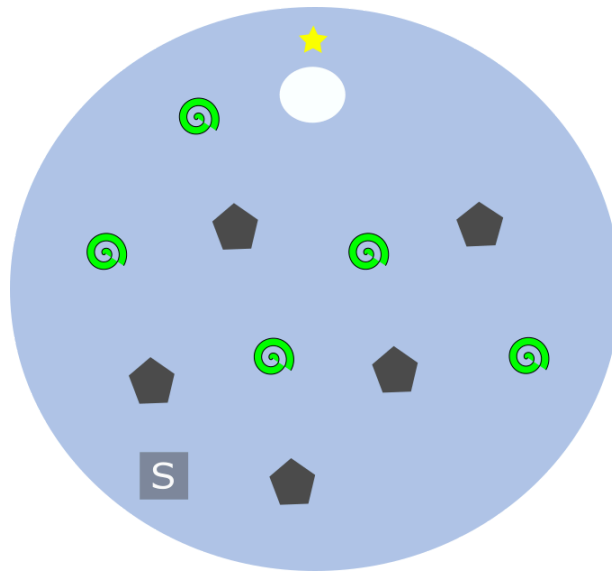
This second procedure has the entire open-field tank rotated 90 degrees from its prior orientation in Procedure 1 to test if the bobtails utilize cues external to the tank such as light and the polarization of light:



If the times and path lengths initially increase or differ from the baseline values received from the learning curve produced from P1, we can conclude that the bobtails may be utilizing external tank cues (such as light position or polarization) to navigate.

Experimental Procedure 3 (Investigation of Path Integration)

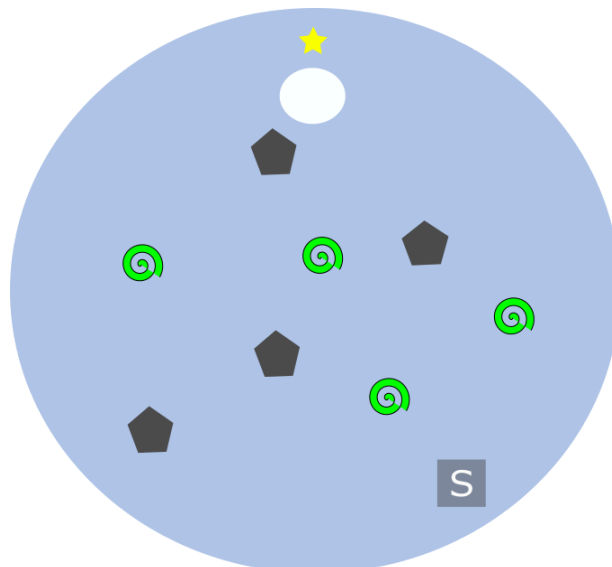
Here, the bobtails are placed in the designated starting position which is now shifted 90 degrees from the prior starting position:



If the times and path lengths increase or differ from the baseline values received from the learning curve produced from P1 after the rotation, we can conclude that the bobtails may be utilizing a kind of path-integrating self cue to navigate.

Experimental Procedure 4 (Investigation of Piloting Navigation)

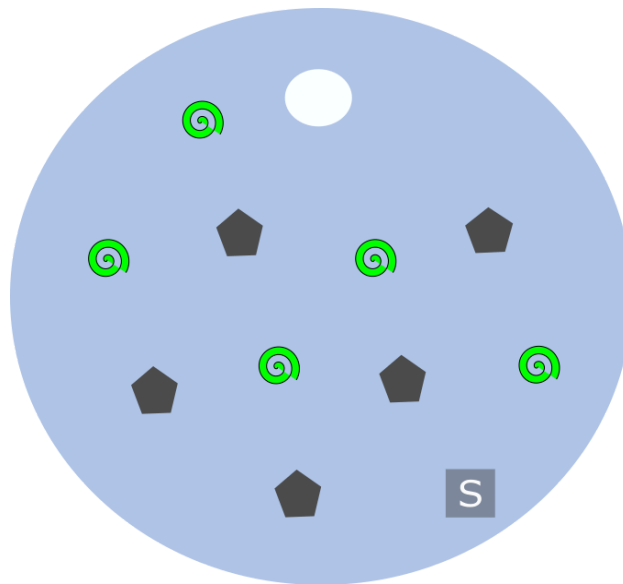
Bobtails are placed in the designated starting position as in Procedure 1 after the objects inside the tank are manipulated haphazardly: some are taken out, moved around, etc. The in-tank manipulation will occur once at the start of the testing procedure:



If the bobtails' escape times or path lengths increase or deviate from baseline P1 values when internal tank cues are altered, this suggests reliance on piloting navigation in escaping.

Experimental Procedure 5 (Investigation of Beacons Navigation)

Bobtails are placed in the designated starting position as in Procedure 1 after the landmark over the hole is removed:



If the bobtails' escape times or path lengths increase or deviate from baseline after removal of the hole marker, this suggests that they are utilizing beacons as a navigational strategy.

Behavioral Contingency Plans

As this is a behavioral experiment, there are a multitude of issues that can arise. The contingency plans for perceived experimental issues are as follows: If the bobtails navigate immediately to the hole with no apparent learning period, the experimental tank

can be fitted with more holes, introducing a kind of decoy-hole procedure where multiple holes can be open or closed in between trials. In the event that the bobtails cannot learn to escape the experimental tank, multiple changes can be put into effect like making the experimental tank itself smaller, drilling a larger escape hole, and/or trying to reduce the stress load of the individuals (separate them out into individual tanks, reduce handling time, etc).

Data Analysis and Interpretation

Distance travel by the bobtails to escape the open field maze will be measured using ImageJ.

GoPro measurements of both time and distance from Procedure 1 will be analyzed with linear regression (trial number vs. escape time, ANOVA) to determine if there is a learning curve present. A significant ANOVA and negative slope of the line produced would indicate that the bobtails are truly learning. From here, a breakpoint analysis (Davies test) will be performed on this same dataset to find if there is a breakpoint or leveling-off phase in the slope; this is done by fitting two linear regressions to the data and finds the break in the 2-phase slope- everything after this breakpoint serves as the learning plateau.

Measurements of time and path length from Procedures 2-5 will be analyzed using a mixed-effects Cox proportional hazards model, a form of survival analysis, which is a type of statistical analysis where the outcome measured is the time (or other length-type variable, like distance traveled) before an event occurs, and can also account for repeated measures. The event, in this case, would be the bobtail escaping the experimental tank. This type of analysis is also helpful when not all of your sample subjects experience the event, which leads to data censoring. Censoring would occur when some individuals take more than the allotted 10 minutes to find and escape the

hole (or never escape). Thus, the data collected would be censored, meaning its true value (if the experiment continued infinitely long) is not known. Experiments 2-5 test various cues the bobtails might use to navigate to the exit. Data from the first trial of each of those experiments (time to escape and distance traveled to escape) will be compared by the above survival analysis to the post-breakpoint data from the previous trial. Significantly longer times or distances from Procedures 2-5 would indicate that the bobtails likely use the associated cues for navigation.

Data Management Plan

During data collection, all raw video files from the GoPro camera (estimated at ~2–4 GB per hour of footage) and measurement data (e.g., time to escape, path length) exported from ImageJ will be stored in three locations for redundancy:

1. The primary research computer's internal drive.
2. An external solid-state drive (SSD) dedicated to data backup.
3. WWU OneDrive to ensure off-site redundancy.

Following analysis, all processed data, R scripts, statistical outputs, and summarized datasets (such as .CSV files for path lengths and escape times) will also be stored in these three locations.

For long-term storage, all raw and processed data will be archived in Zenodo, the lab's preferred data repository. This ensures long-term preservation and accessibility of the dataset.

Estimated total storage needs:

Raw video data: ~100–150 GB

Processed image analysis & summary data: ~1–2 GB

Statistical scripts and documentation: < 1 GB

Total estimate: ~160 GB

The existing lab infrastructure (external hard drives and institutional cloud storage) is sufficient for short-term needs. Zenodo offers free public archiving of research datasets up to 50 GB per upload (and larger upon request), so no additional repository costs are anticipated.

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BUDGET AND BUDGET JUSTIFICATION

Table for budget

- Home Depot items: \$101.06
 - Sump bin
 - Tubing
- Canister filter: \$229.99
- Misc petsmart items: \$212.90
 - Sand
 - Filters
 - Heaters
 - Water quality equipment
- Experimental tank: ~\$300-400
- Squids: \$120
- Mysids: \$1,079.64
- Shrimp: \$381.09
- Protein skimmer: \$339.99

Total: \$2864.67

Most experimental items and tanks along with their respective aquarium equipment are all entirely reusable for projects down the line. The food for the bobtails, however, is the biggest source of fund usage. During the first ~40 days of their lives, the bobtails were feeding on mysid shrimp, a small shrimp shipped over from Florida. These shrimp are notoriously hard to culture over long periods of time, and we saw the same difficulty when it came time to keep placing repeat orders. Another factor in the high mysid shrimp cost was the shipping amount- getting overnight shipping from Florida was unfortunately not cheap. Once the squid were transitioned onto their adult food, grass shrimp, culturing their food became a lot easier as these shrimp are much hardier and last a lot longer. On this note, moving operations to Rosario will save money on ordering food by getting shrimp from the ocean ourselves and on Instant Ocean by not needing to fabricate tons of seawater.